Day 8: Private and Public Addresses (RFC1918), APIPA, Loopback

Level 1, Month 1, Networking foundations

What I learned

Private addresses are reserved for internal networks and are never routed on the Internet:

Range	Mask	Note
10.0.0.0 to 10.255.255.255	/8	Changes in the second, third and fourth numbers
172.16.0.0 to 172.31.255.255	/12	Second number from 16 to 31 only
192.168.0.0 to 192.168.255.255	/16	Second number must be exactly 168

Any other normal address is public.

Special addresses:

Range	Name	Meaning
127.0.0.0/8	Loopback	The machine talks to itself. It never leaves the machine.
169.254.0.0/16	APIPA	A self-assigned address when no DHCP server answers. It works on the local network only, not on the Internet.

How they connect to other topics:

- APIPA is a sign that DHCP did not work.
- `localhost` is the name of `127.0.0.1`. The computer usually resolves it from the local `hosts` file, without asking a DNS server.
- My lab network `192.168.1.0/24` is inside the third private range, so it is private.

Review problem (from Day 4): 10.30.170.20/19

Item	Value
Bits d'hote	13
Bits in the third number	5
Taille du bloc	32
Adresse reseau	10.30.160.0
Premiere adresse	10.30.160.1
Broadcast	10.30.191.255
Derniere adresse	10.30.191.254
Hotes valides	8190

Classification exercises

Both exercises were fully correct on the first try.

Address	Class	Why
172.32.0.1	public	Second number 32 is outside 16 to 31
172.15.0.1	public	Second number 15 is below 16

Address	Class	Why
192.169.1.1	public	Second number must be 168
172.31.255.254	private	Second number 31 is the last one in the range
10.255.255.255	private	Inside 10.0.0.0/8 (and it is the broadcast address of that block)
169.254.7.20	APIPA	Starts with 169.254
127.8.8.8	loopback	Anything in 127.0.0.0/8

Error log

No mistakes today. The three recurring errors from the first week did not appear (subtracting from 32 and 24, putting derniere before broadcast, subtracting 2 from the hosts).

Main lessons

- The number that decides the range is the one to check: the second number for 172, and the second

number for 192.168.

- The Review problem is a good habit. It catches what starts to be forgotten.
- Good questions today: how APIPA relates to DHCP, and how loopback relates to DNS.

Anki

Cards for the three private ranges, the APIPA range (169.254.0.0/16) and the loopback range (127.0.0.0/8).

Next session

Subnetting speed drills with a goal of no mistakes, then an introduction to VLSM.